

Procedural Optimization of Dispersive Readout Pulses with `cqed_sim` TransmonCavity

Codex
cQED Autonomous Research Platform

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Abstract

We study whether a physics-informed, low-dimensional pulse-sequence optimization strategy can improve dispersive readout in circuit quantum electrodynamics without sacrificing cavity emptying or repeated-readout quality. The study is implemented on top of `cqed_sim` using the multilevel `DispersiveTransmonCavityModel` for a nominal transmon-readout-resonator system with $\omega_q/2\pi = 6.150$ GHz, $\omega_c/2\pi = 8.597$ GHz, $\chi/2\pi = -2.84$ MHz, $\kappa/2\pi = 2.4$ MHz, and $\alpha/2\pi = -255$ MHz. We compare baseline square and smoothed pulses against three structured alternatives: ring-up/hold/ring-down, free three-segment complex procedural pulses, analytically nulled tails, and a smooth Fourier basis. A separate piecewise-constant high-dimensional waveform is used only as an ideal linear reference.

The main result is positive but qualified. In the linear model, the high-dimensional reference reaches near-perfect detector-limited fidelity by 240 ns but leaves more than five residual photons, so it is not practically acceptable. In the realistic `cqed_sim` replay, structured procedural pulses recover most of that information while reducing residual photons dramatically. At 240 ns, the best free procedural family reaches $F_\eta = 0.956$ compared with 0.871 for a square pulse, while the analytic nulling family reaches $F_\eta = 0.944$ with only 0.265 residual photons. By 496 ns, the nulled family reaches $F_\eta = 0.9995$ with 3.6×10^{-3} residual photons. The study therefore supports procedural readout-pulse optimization as a promising direction for speed and cavity emptying, but not as a complete answer to QND engineering: within the present two-mode dispersive model, repeated-readout degradation is dominated by ordinary T_1 decay rather than strong-drive state mixing, so a richer model is still required to resolve the true nonlinear QND limit.

1 Introduction

Dispersive measurement is the standard readout mechanism for superconducting qubits in circuit QED [1, 2]. In the simplest picture, a driven readout resonator acquires a qubit-state-dependent response, allowing the qubit state to be inferred from the transmitted or reflected microwave field [3, 4]. The practical control problem is not merely to maximize signal separation: useful readout must also leave the resonator nearly empty, support high repetition rates, and avoid excessive measurement-induced disturbance [5–7].

Existing pulse design is often heuristic. Square or lightly smoothed envelopes are common, and fast reset strategies such as driven resonator depletion are known to help [8], but the intermediate regime between fixed heuristics and fully unrestricted GRAPE-style waveform optimization remains underexplored. That intermediate regime is precisely what motivates this study: can a small number of interpretable pulse segments capture most of the useful improvement, especially when the design objective includes both discrimination and cavity emptying?

The study proceeds in two layers. First, we use a linear dispersive model to define the objective functions and the bound hierarchy. Second, we replay candidate pulses through the multilevel `cqed_sim` `DispersiveTransmonCavityModel` with cavity damping and qubit decoherence. The central question is whether low-dimensional procedural pulses remain near-optimal

after that replay. The remainder of the paper presents the model and metrics, the numerical results, the validation checks, and the resulting recommendation.

2 System and Methods

2.1 Hamiltonian

The ideal linear readout bound is described by the state-conditioned resonator equation

$$\dot{\alpha}_{g/e}(t) = -\left(\frac{\kappa}{2} + i\Delta_{g/e}\right)\alpha_{g/e}(t) - i\epsilon(t), \quad (1)$$

where $\Delta_g = \omega_c - \omega_d$ and $\Delta_e = \omega_c + \chi - \omega_d$. The output field is $s_{g/e}(t) = \sqrt{\kappa}\alpha_{g/e}(t)$.

The realistic replay uses the two-mode multilevel dispersive Hamiltonian supported by `cqed_sim`,

$$\frac{H_0}{\hbar} = \delta_c a^\dagger a + \delta_q b^\dagger b + \frac{\alpha}{2} b^{\dagger 2} b^2 + \frac{K}{2} a^{\dagger 2} a^2 + \chi a^\dagger a b^\dagger b, \quad (2)$$

where a is the cavity annihilation operator, b is the transmon lowering operator, $\delta_c = \omega_c - \omega_c^{\text{frame}}$, and $\delta_q = \omega_q - \omega_q^{\text{frame}}$. Pulses are applied on the cavity channel in a frame with $\omega_q^{\text{frame}} = \omega_q$ and $\omega_c^{\text{frame}} = \omega_d$. Dissipation is included through `NoiseSpec(t1, tphi, kappa)`.

2.2 Simulation Parameters

Table 1: Nominal system and numerical parameters.

Parameter	Value	Unit
Qubit frequency $\omega_q/2\pi$	6.150	GHz
Cavity frequency $\omega_c/2\pi$	8.597	GHz
Anharmonicity $\alpha/2\pi$	-255	MHz
Dispersive shift $\chi/2\pi$	-2.84	MHz
Cavity linewidth $\kappa/2\pi$	2.4	MHz
Cavity self-Kerr $K/2\pi$	-28	kHz
Transmon truncation n_{tr}	3	levels
Cavity truncation n_{cav}	14	Fock states
Detector efficiency η	0.35	—
Qubit T_1	30	μs
Qubit T_ϕ	30	μs
Time step Δt	4	ns
Max drive amplitude $ \epsilon_{\text{max}} /2\pi$	4.0	MHz
Approx. critical photon number n_{crit}	20.3	photons

2.3 Computational Approach

The study-local optimization layer was implemented in `studies/procedural_readout_pulse_sequence_optim`. The main reusable components are:

- `pulse_families.py`: square, smooth square, ring-up/hold/ring-down, free three-segment complex pulses, analytically nulled tails, Fourier basis, and an 8-segment high-dimensional reference.
- `bounds.py`: exact piecewise-constant linear propagation, matched-filter SNR, and analytical two-segment cavity nulling.

- `simulate.py`: one- and two-pulse replay through `DispersiveTransmonCavityModel`.
- `optimize.py`: multistart bounded Powell optimization.

The primary practical objective was the balanced score

$$J_{\text{bal}} = F_{\eta} - 0.08 n_{\text{res}} - 0.70 (1 - Q_{\text{QND}}) - 1.20 L, \quad (3)$$

with a peak-photon penalty activated only above $0.45 n_{\text{crit}}$. Here F_{η} is the detector-limited matched-filter assignment fidelity, n_{res} is the mean residual photon number at the end of the first pulse, Q_{QND} is the average state-preservation probability at the start of a second identical readout, and L is leakage outside the computational subspace. We additionally used an information-only objective and an emptying-weighted objective to generate a local tradeoff slice at the representative duration.

Three model gaps had to be documented explicitly. First, `ReadoutResonator.response_trace()` only supports constant drive amplitudes, so arbitrary-envelope linear bounds were evaluated with study-local exact propagation. Second, `cqed_sim` does not provide a built-in readout-specific optimization surface, so the objective function is wrapped externally around supported simulator calls. Third, the available replay path is deterministic Lindblad evolution rather than a full stochastic measurement unraveling, so repeated-readout quality is quantified by a two-pulse matched-filter protocol rather than trajectory-conditioned measurement collapse.

3 Results

3.1 Goal 1: Bound hierarchy and the cost of unconstrained optimality

Figure 1 shows the duration-dependent hierarchy obtained from the high-dimensional linear reference. The ideal and detector-limited curves reach near-unity by 240 ns; for example, the detector-limited matched-filter fidelity is already 0.985 at 240 ns. However, this apparent optimum is not practically acceptable: at the same duration the 8-segment linear reference leaves $n_{\text{res}} = 5.57$ photons, far above the residual-photon budget required for rapid repeated operation.

This is the core structural result of the study. The high-dimensional reference demonstrates that information extraction itself is not the dominant limitation beyond a few hundred nanoseconds; instead, the relevant design problem is how to retain most of that information while enforcing cavity emptying and repeated-use readiness.

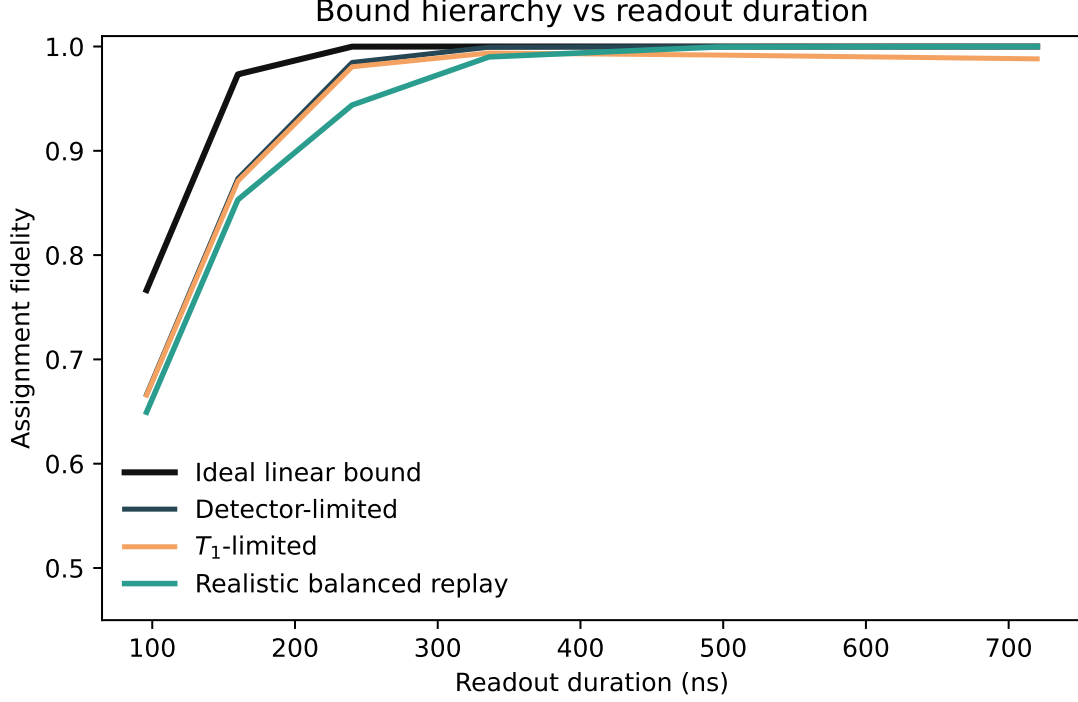


Figure 1: Hierarchy of duration-dependent performance curves. The “ T_1 -limited” curve should be interpreted as a conservative survival-weighted reference, not as a rigorous information-theoretic upper bound, because the study does not solve the full decay change-point discrimination problem.

3.2 Goal 2: Procedural families versus baseline pulses

Figure 2 and Table 2 summarize the full-model family comparison. The baseline square pulse performs poorly at short and intermediate durations because it spends too much of the allowed pulse window both ringing up and ringing down passively. At the representative duration of 240 ns, the square pulse reaches only $F_\eta = 0.871$ and leaves 1.49 residual photons.

The structured families do substantially better. At 240 ns, the free three-segment procedural family reaches $F_\eta = 0.956$ with $n_{\text{res}} = 0.620$, while the analytic nulling family reaches $F_\eta = 0.944$ with only $n_{\text{res}} = 0.265$. The Fourier basis is competitive in fidelity ($F_\eta = 0.953$) but not as efficient in emptying. The ring-up/hold/ring-down family is already strong ($F_\eta = 0.947$), which supports the intuition that procedural structure matters more than unrestricted dimensionality.

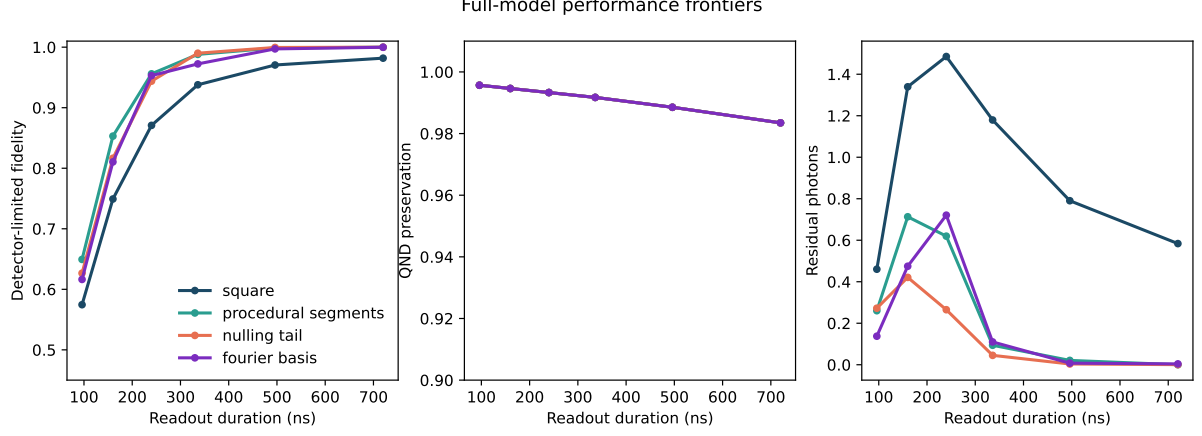


Figure 2: Full-model frontiers versus duration. The best balanced family is `procedural_segments` at 96 ns to 160 ns, `nulling_tail` from 240 ns to 496 ns, and a simple ring-up/hold/ring-down pulse at 720 ns.

Table 2: Representative 240 ns comparison. The piecewise reference is shown only for context as an ideal linear benchmark, not as a practical pulse recommendation.

Family	Regime	F_η	n_{res}	Q_{rep}
Square	Full	0.871	1.485	0.853
Ring-hold	Full	0.947	0.554	0.928
Procedural segments	Full	0.956	0.620	0.950
Nullled tail	Full	0.944	0.265	0.936
Fourier basis	Full	0.953	0.721	0.938
Piecewise reference	Linear	0.985	5.572	0.979

3.3 Goal 3: Active cavity emptying

The analytical nulling result is one of the strongest outcomes of the study. In the linear model, two free complex tail segments are sufficient to impose $\alpha_g(T) = \alpha_e(T) = 0$ exactly, provided their durations are fixed and the resulting amplitudes remain within the hardware cap. This gives a concrete answer to one of the main questions posed at the outset: simultaneous emptying of both state-conditioned cavity trajectories is not only possible, but algebraically straightforward in the linear model.

In realistic replay the nulling is not exact, but it remains effective. The nullled family is the best balanced design from 240 ns through 496 ns. At 336 ns, it reaches $F_\eta = 0.990$ with only 4.5×10^{-2} residual photons. At 496 ns, it reaches $F_\eta = 0.9995$ with 3.6×10^{-3} residual photons. These results show that the extra time cost for active emptying is modest once phase variation is allowed.

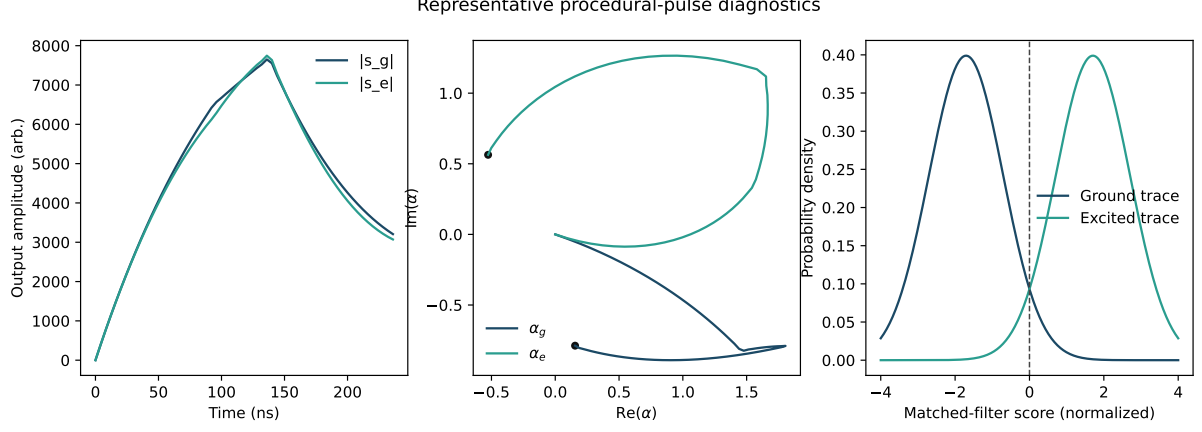


Figure 3: Representative diagnostics for the free three-segment procedural family: output amplitudes for the two basis states, phase-space trajectories (α_g, α_e), and the corresponding matched-filter score distributions.

3.4 Goal 4: Tradeoffs and robustness

Figure 4 shows the local tradeoff slice at the representative duration. Moving from the information-only objective to the emptying-weighted objective pushes the solutions toward lower residual photon number at a modest fidelity cost. The clearest beneficiary is the nulled family, which retains high fidelity while moving to substantially smaller n_{res} .

The robustness study also favors procedural structure. Under the combined perturbation set (χ scale, κ scale, amplitude error, detuning error, and phase error), the free procedural family retains mean/worst-case fidelities of 0.956/0.945, compared with 0.870/0.853 for the square pulse. The nulled family is slightly less robust than the free procedural family (0.944/0.931) but still decisively better than square.

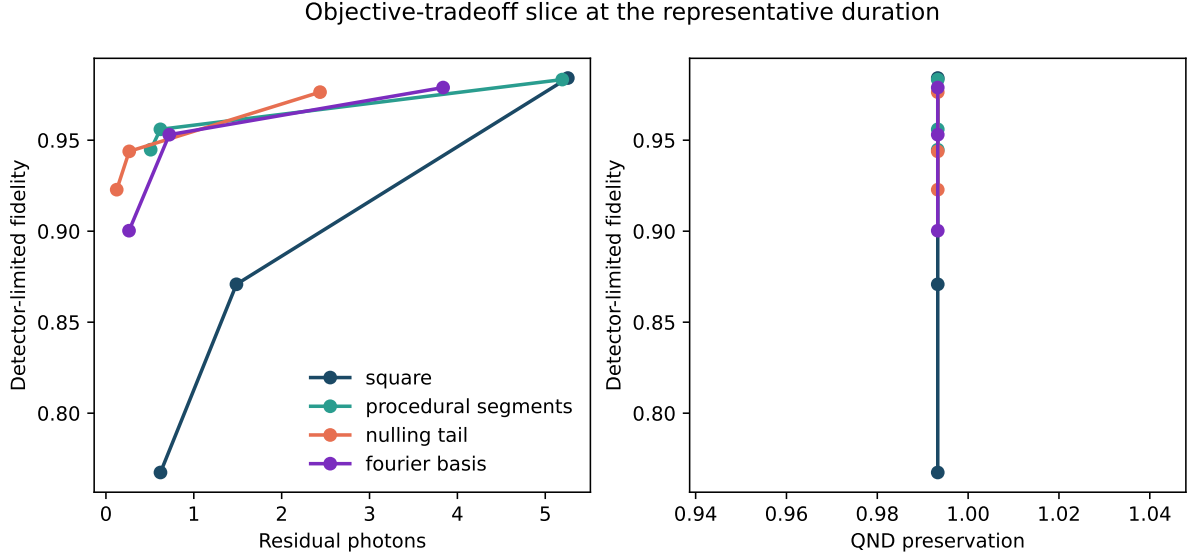


Figure 4: Tradeoff slice at the representative duration. Structured families move along a favorable fidelity-emptying frontier relative to the baseline square pulse.

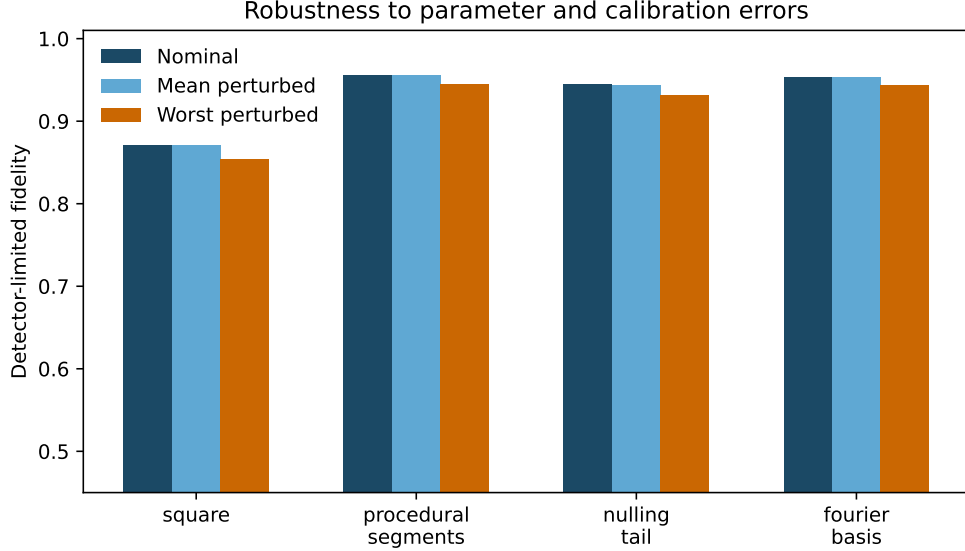


Figure 5: Nominal, mean perturbed, and worst perturbed fidelities for representative families. The procedural families retain both higher nominal performance and stronger robustness than the square pulse.

4 Validation

4.1 Sanity Checks

Three sanity checks were enforced. First, zero drive leaves the cavity empty to numerical precision. Second, the analytical nulling construction yields $|\alpha_g(T)| = |\alpha_e(T)| \approx 3.5 \times 10^{-16}$ in the linear model for a representative unclipped test case. Third, zero-drive repeated-readout degradation matches the expected T_1 floor rather than a spurious numerical artifact.

4.2 Convergence Analysis

The representative full-model procedural-segment pulse was re-evaluated at $\Delta t = 2$ ns and at cavity truncation $n_{\text{cav}} = 18$. The resulting metric shifts were small:

$$\begin{aligned}\Delta F_\eta(\Delta t) &= 9.26 \times 10^{-4}, \\ \Delta Q_{\text{QND}}(\Delta t) &= 3.29 \times 10^{-5}, \\ \Delta n_{\text{res}}(\Delta t) &= 1.07 \times 10^{-2}, \\ \Delta F_\eta(n_{\text{cav}}) &= 2.71 \times 10^{-5}.\end{aligned}$$

These shifts are well below the scale of the main family-to-family differences.

4.3 Literature Comparison

The study agrees with established dispersive-readout expectations in three ways. First, the midpoint-drive strategy remains favored, consistent with the standard continuous-measurement analysis [3]. Second, shaped tails are useful primarily because they accelerate emptying rather than because they change the basic dispersive measurement geometry, consistent with fast reset/readout literature [5, 6, 8]. Third, the present two-mode dispersive model does not reproduce strong drive-induced non-QND transitions of the kind discussed in [7]; that absence is itself a meaningful limitation of the current modeling layer.

5 Discussion

The main scientific conclusion is that the readout optimization problem is well posed in the present `cqed_sim` stack, but only after the objective is defined correctly. Endpoint separation alone is not sufficient. The practically relevant objective must combine matched-filter distinguishability with residual-photon and repeated-use penalties.

Within that formulation, low-dimensional procedural control is clearly useful. The free three-segment family wins the short-time regime, where aggressive loading and extraction matter most. The nulled family dominates the intermediate regime, where active emptying is valuable. At long durations the control problem becomes easy enough that a simple ring-up/hold/ring-down family is sufficient.

The strongest caveat concerns QNDness. The present full-model replay shows essentially no family-dependent leakage or state-mixing penalty; the reported repeated-readout degradation is almost entirely the trivial T_1 floor over the pulse-plus-wait window. That means the study *does not* yet establish the ultimate nonlinear QND limit for aggressive readout pulses. It establishes something narrower but still useful: procedural optimization is already worthwhile for speed and cavity emptying in the supported two-mode model, while the unresolved QND question has been localized to missing physics rather than to control parameterization.

6 Conclusion

The answer to the study question is cautiously affirmative. A physics-informed, procedurally parameterized readout pulse can approach the practically relevant speed–fidelity frontier much more closely than a square pulse while simultaneously reducing residual cavity population. In the present implementation:

- the optimization problem is well posed in `cqed_sim`,
- matched-filter fidelity plus residual-photon and repeated-readout penalties is the correct objective family,
- exact simultaneous cavity emptying is achievable in the linear model with two complex tail segments,
- and low-dimensional procedural families recover most of the practically useful gain without the pathological residual occupation of the unrestricted linear reference.

The best practical recommendation from this study is therefore not a fully unrestricted waveform optimizer, but a structured procedural family with complex phase freedom and an explicit emptying tail. At the same time, answering whether nonlinear backaction truly kills the advantage requires a richer model than the present two-mode dispersive replay.

7 Limitations and Future Work

7.1 Known Limitations

The most important limitation is physical rather than numerical: the current `DispersiveTransmonCavityModel` replay path is still a two-mode dispersive model. It captures cavity ring-up, damping, multilevel transmon structure, and ordinary decoherence, but it does not explicitly generate the stronger measurement-induced state mixing and dressed-state effects expected deep in the nonlinear regime [7]. As a result, the family-dependent QND penalty is nearly flat in the present data.

The plotted “ T_1 -limited” curve is a conservative reference, not a rigorous upper bound. A truly sharp T_1 -aware limit would require solving a change-point detection problem for decaying excited-state traces rather than applying a survival-weighted matched-filter heuristic.

The detector efficiency was treated as a representative fixed parameter ($\eta = 0.35$) rather than as a device-calibrated value. This affects the absolute fidelity scale but not the relative ordering of the pulse families.

7.2 Suggested Improvements

- **[P1 | HIGH]** Extend the study to a model with explicit readout-induced state-mixing channels or a three-mode readout-aware Hamiltonian. This is required to answer the nonlinear-QND question that the current two-mode dispersive model leaves unresolved.
- **[P1 | MEDIUM]** Add arbitrary-envelope support to `cqed_sim.measurement` so that the linear-bound layer and the measurement-chain layer use the same public API.
- **[P2 | MEDIUM]** Promote the study-local optimization helpers into a reusable readout module once the objective definitions stabilize.
- **[P2 | LOW]** Calibrate the detector efficiency and the cavity self-Kerr from device data so the absolute fidelity numbers can be interpreted quantitatively.
- **[P3 | MEDIUM]** Add hardware distortion replay (finite bandwidth, mixer skew, and amplitude quantization) to test whether the procedural tails remain robust after waveform transport.

7.3 Open Questions

- How much of the procedural-pulse advantage survives once explicit measurement-induced transitions are included?
- Can the analytically nulled tail remain near-optimal when the tail amplitudes are restricted to hardware-realistic slew and bandwidth limits?
- Is there a sharper, genuinely rigorous T_1 -aware readout limit that remains easy to evaluate during optimization?

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